

R basics

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Directory

To know the working directory

```
getwd()
```

```
[1] "G:/Lesson/Done/Research/R Language/R-Language/Basic"
```

Set the working directory

```
#setwd("G\\...\\...\\like-this.csv")
```

Show the file in the working directory

```
list.files()
```

```
[1] "R-basics.rmarkdown" "R basics.qmd"      "R basics.rmarkdown"
```

Conditional Sentence

```
a <- 10
b <- 12
if(a>b)
{print("a is greater than b")} else if(a==b)
{print("a and b are equal")} else
{print("b is greater than a")}
```

```
[1] "b is greater than a"
```

Loop

```
for(x in 1:10)
{print(x)}
```

```
[1] 1
[1] 2
[1] 3
[1] 4
[1] 5
[1] 6
[1] 7
[1] 8
[1] 9
[1] 10
```

Function

```
avarage <- function(x,y,z){  
ava=(x+y+z)/2  
print(ava)  
}  
  
avarage(2,3,4)
```

[1] 4.5

Data Type

```
a <- 10  
typeof(a)
```

[1] "double"

```
b <- "swarup"  
typeof(b)
```

[1] "character"

```
c <- TRUE  
typeof(c)
```

[1] "logical"

```
d <- 10L  
typeof(d)
```

[1] "integer"

```
e <- 5i  
typeof(e)
```

[1] "complex"

Sub Setting

```
number <- 1:100
```

```
odd_number <- seq(1,99,2)  
from100_1 <- seq(100,1,-1)
```

```
from100_1
```

```
[1] 100 99 98 97 96 95 94 93 92 91 90 89 88 87 86 85 84 83  
[19] 82 81 80 79 78 77 76 75 74 73 72 71 70 69 68 67 66 65  
[37] 64 63 62 61 60 59 58 57 56 55 54 53 52 51 50 49 48 47  
[55] 46 45 44 43 42 41 40 39 38 37 36 35 34 33 32 31 30 29  
[73] 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11  
[91] 10 9 8 7 6 5 4 3 2 1
```

```
odd_number
```

```
[1] 1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35 37 39 41 43 45 47 49  
[26] 51 53 55 57 59 61 63 65 67 69 71 73 75 77 79 81 83 85 87 89 91 93 95 97 99
```

```
odd_number[10:20]
```

```
[1] 19 21 23 25 27 29 31 33 35 37 39
```

```
odd_number[c(12,32,50,99)]
```

```
[1] 23 63 99 NA
```

```
odd_number>20
```

```
[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE  
[13] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE  
[25] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE  
[37] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE  
[49] TRUE TRUE
```

Variable type change

```
double <- 3.1416  
typeof(double)
```

```
[1] "double"
```

```
int <- as.integer(double)  
int
```

```
[1] 3
```

```
typeof(int)
```

```
[1] "integer"
```

```
srt <- as.character(double)  
srt
```

```
[1] "3.1416"
```

```
typeof(srt)
```

```
[1] "character"
```

```
logi <- as.logical(double)  
logi
```

```
[1] TRUE
```

```
typeof(logi)
```

```
[1] "logical"
```

Creating Dataset in R

Simple

```
student_data <- data.frame (name=c("shadhin","joy","durjoy"),roll=c(23,45,32))
```

Creating with rep function

here 'rep' means repeated This I will use in the One Way ANOVA

```
data_rep <- data.frame(method=rep(c("No 1","No 2","No 3"),each=4), marks=c(1,2,3,4,5,6,7,8,9,10,11,12))
```

	method	marks
1	No 1	1
2	No 1	2
3	No 1	3
4	No 1	4
5	No 2	5
6	No 2	6
7	No 2	7
8	No 2	8
9	No 3	9
10	No 3	1
11	No 3	2
12	No 3	3

With matrix

```
met <- matrix(c(23,43,24,24,23,34,32,35,32,35,35,35,34,NA,36), nrow=5,ncol=3,byrow=TRUE)
```

	[,1]	[,2]	[,3]
[1,]	23	43	24
[2,]	24	23	34
[3,]	32	35	32
[4,]	35	35	35
[5,]	34	NA	36

```
is.data.frame(met)
```

```
[1] FALSE
```

```
length(met)
```

```
[1] 15
```

Read file and Some data exploration

```
Family_name <-c("Durjoy", "Swapon", "Lipy")
Height <- c(155,150,120) #in cm
Weight <- c(55,65,45) #in kg

#Making it a data set or data frame
Family_info <- data.frame(Family_name, Height, Weight)

# dataset view
Family_info
```

	Family_name	Height	Weight
1	Durjoy	155	55
2	Swapon	150	65
3	Lipy	120	45

```
print(is.data.frame(Family_info))
```

```
[1] TRUE
```

```
#number of row
print(nrow(Family_info))
```

```
[1] 3
```

```
#number of colulmn
print(ncol(Family_info))
```

```
[1] 3
```

```
#first six row or just the snipit
head(Family_info)
```

	Family_name	Height	Weight
1	Durjoy	155	55
2	Swapon	150	65
3	Lipy	120	45

```
#Last six row
tail(Family_info)
```

	Family_name	Height	Weight
1	Durjoy	155	55
2	Swapon	150	65
3	Lipy	120	45

```
#summary of the dataset
summary(Family_info)
```

Family_name	Height	Weight
Length:3	Min. :120.0	Min. :45
Class :character	1st Qu.:135.0	1st Qu.:50
Mode :character	Median :150.0	Median :55
	Mean :141.7	Mean :55
	3rd Qu.:152.5	3rd Qu.:60
	Max. :155.0	Max. :65

```
#column name
names(Family_info)
```

```
[1] "Family_name" "Height"      "Weight"
```

```
#showing one column
Family_info$Family_name
```

```
[1] "Durjoy" "Swapon" "Lipy"
```



```
#showing the length of farmer data , how many column are there  
length(Family_info)
```

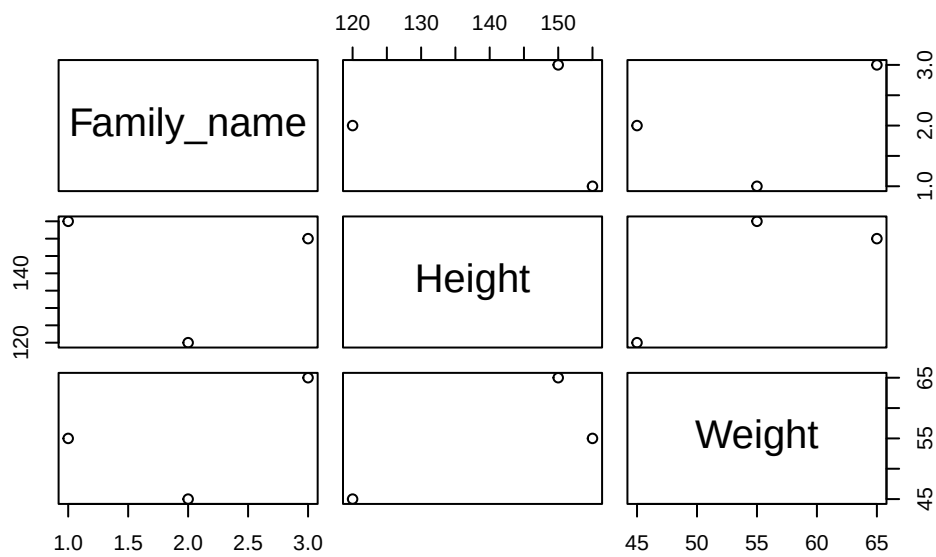
```
[1] 3
```

```
#length of the column  
length(Family_info$Height)
```

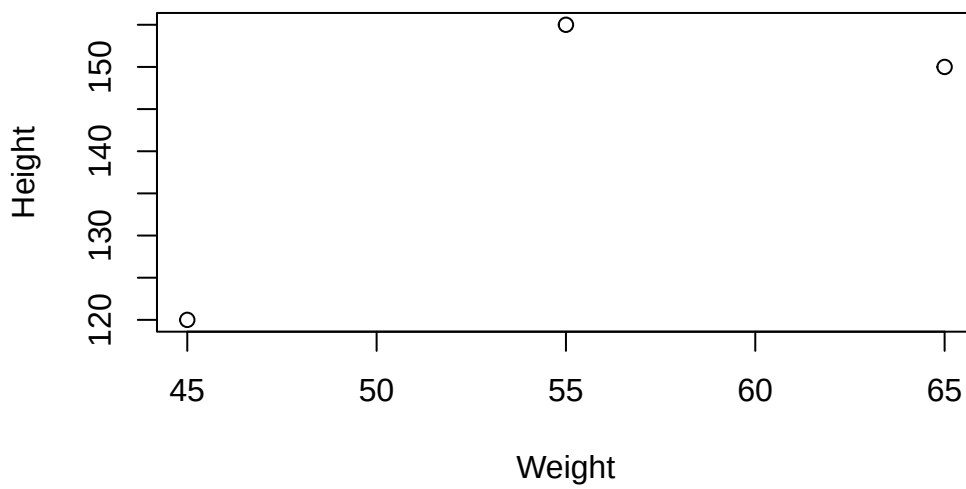
```
[1] 3
```

Practice 1 - BMI Calculation

```
#column info  
Family_name <-c("Durjoy", "Swapon", "Lipy")  
Height <- c(155,150,120) #in cm  
Weight <- c(55,65,45) #in kg  
  
#Making it a data set or data frame  
Family_info <- data.frame(Family_name, Height, Weight)  
  
#Some visualization  
plot(Family_info)
```



```
plot(Height~Weight, Family_info)
```



```
#data filter***
Family_info[Family_info$Height==155,]
```

	Family_name	Height	Weight
1	Durjoy	155	55

```
# Creating a new column of BMI
Family_info$BMI <-Family_info$Weight/(Family_info$Height*0.01)^2
Family_info
```

	Family_name	Height	Weight	BMI
1	Durjoy	155	55	22.89282
2	Swapon	150	65	28.88889
3	Lipy	120	45	31.25000

```
# Adding a new category***
Family_info$Health_status <- ifelse(Family_info$BMI<=18.5,"Underweight",ifelse(Family_info$BMI>18.5,"Overweight",ifelse(Family_info$BMI>30,"Obesity","Normal Weight")))
Family_info
```

	Family_name	Height	Weight	BMI	Health_status
1	Durjoy	155	55	22.89282	NORMAL WEIGHT
2	Swapon	150	65	28.88889	OVERWEIGHT
3	Lipy	120	45	31.25000	OBESITY